

Name: _____

FRACTIONS ALL OPERATIONS, (DIVIDING FRACTIONS)

Date: _____

LO: I can divide fractions by flipping the second fraction and multiplying.

Dividing fractions

To divide by a fraction, we keep, change, flip: keep the first fraction, change division to multiplication, and flip (find the reciprocal of) the second fraction.

Remember: The reciprocal of $\frac{a}{b}$ is $\frac{b}{a}$. Dividing by a fraction makes the answer bigger.

Quick examples

●	$\frac{2}{3} \div \frac{1}{4} = \frac{8}{3}$	flip and multiply
●	$\frac{3}{5} \div \frac{2}{7} = \frac{21}{10}$	keep, change, flip
●	$\frac{1}{2} \div 3 = \frac{1}{6}$	$3 = \frac{3}{1}$, flip to $\frac{1}{3}$
●	$\frac{2}{3} \div \frac{1}{4} = \frac{2}{12}$	multiplied without flipping
●	$\frac{3}{5} \div \frac{2}{7} = \frac{6}{35}$	did not flip second fraction

Key idea

Keep the first fraction, change the operation to multiplication, flip the second fraction.

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$$

Keep, Change, Flip

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – simple division

Calculate $\frac{2}{3} \div \frac{1}{4}$

Keep $\frac{2}{3}$, change to $\times$, flip $\frac{1}{4}$ to $\frac{4}{1}$

$$\frac{2}{3} \times \frac{4}{1} = \frac{8}{3}$$

$$= 2\frac{2}{3}$$

$$= 2\frac{2}{3}$$

Flip the second fraction then multiply across.

Example 2 – both non-unit

Calculate $\frac{3}{5} \div \frac{2}{7}$

Keep $\frac{3}{5}$, change to $\times$, flip $\frac{2}{7}$ to $\frac{7}{2}$

$$\frac{3}{5} \times \frac{7}{2} = \frac{21}{10}$$

$$= 2\frac{1}{10}$$

$$= 2\frac{1}{10}$$

Multiply numerators and denominators separately.

Example 3 – dividing by a whole number

Calculate $\frac{4}{5} \div 2$

Write 2 as $\frac{2}{1}$

Keep $\frac{4}{5}$, change to $\times$, flip to $\frac{1}{2}$

$$\frac{4}{5} \times \frac{1}{2} = \frac{4}{10} = \frac{2}{5}$$

$$= \frac{2}{5}$$

A whole number is just a fraction with denominator 1.

Example 4 – mixed numbers

Calculate $1\frac{1}{2} \div \frac{3}{4}$

Convert: $\frac{3}{2} \div \frac{3}{4}$

Keep $\frac{3}{2}$, change to $\times$, flip to $\frac{4}{3}$

$$\frac{3}{2} \times \frac{4}{3} = \frac{12}{6} = 2$$

$$= 2$$

Convert mixed numbers to improper fractions first.

YOUR TURN – PRACTICE (deliberate practice)

1. Dividing unit fractions

Use keep, change, flip.

- a) $\frac{1}{2} \div \frac{1}{3}$
- b) $\frac{1}{4} \div \frac{1}{5}$
- c) $\frac{1}{3} \div \frac{1}{6}$
- d) $\frac{1}{5} \div \frac{1}{2}$

2. Non-unit fractions

Divide. Simplify your answer.

- a) $\frac{3}{4} \div \frac{1}{2}$
- b) $\frac{2}{3} \div \frac{4}{5}$
- c) $\frac{5}{8} \div \frac{3}{4}$
- d) $\frac{4}{7} \div \frac{2}{3}$

3. Dividing by whole numbers

Divide each fraction by the whole number.

- a) $\frac{3}{4} \div 3$
- b) $\frac{5}{6} \div 5$
- c) $\frac{2}{3} \div 4$
- d) $\frac{7}{8} \div 7$

4. Mixed numbers

Convert to improper fractions first.

- a) $1\frac{1}{2} \div \frac{1}{4}$
- b) $2\frac{1}{3} \div \frac{2}{3}$
- c) $1\frac{3}{4} \div \frac{7}{8}$
- d) $3\frac{1}{5} \div 4$

5. Mixed division

Divide and simplify.

- a) $\frac{5}{6} \div \frac{2}{3}$
- b) $\frac{9}{10} \div \frac{3}{5}$
- c) $2\frac{1}{2} \div 1\frac{1}{4}$
- d) $\frac{7}{12} \div \frac{7}{6}$
- e) $4 \div \frac{2}{5}$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) To divide by a fraction, flip the second fraction and multiply ☐

Correct answer: _____
Reason: _____

b) $\frac{2}{3} \div \frac{1}{4} = \frac{2}{12}$ ☐

Correct answer: _____
Reason: _____

c) The reciprocal of $\frac{3}{5}$ is $\frac{5}{3}$ ☐

Correct answer: _____
Reason: _____

d) Dividing by a fraction always gives a smaller answer ☐

Correct answer: _____
Reason: _____

e) $\frac{1}{2} \div 3 = \frac{1}{6}$ ☐

Correct answer: _____
Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A ribbon is $\frac{3}{4}$ m long. It is cut into pieces that are each $\frac{1}{8}$ m long. How many pieces can be cut? If each piece costs 25p, what is the total cost?

Expression: _____

Simplified: _____

b) A jug holds $2\frac{1}{2}$ litres of juice. Each glass holds $\frac{1}{3}$ litre. How many full glasses can be poured? How much juice is left over?

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

